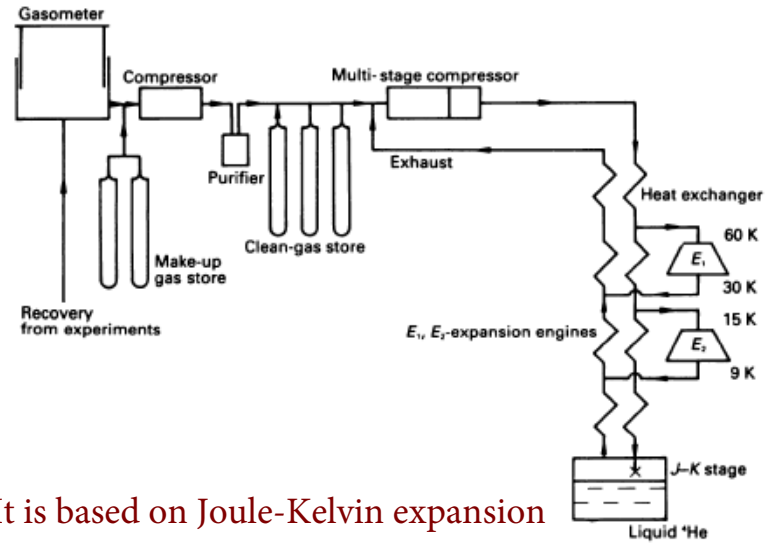
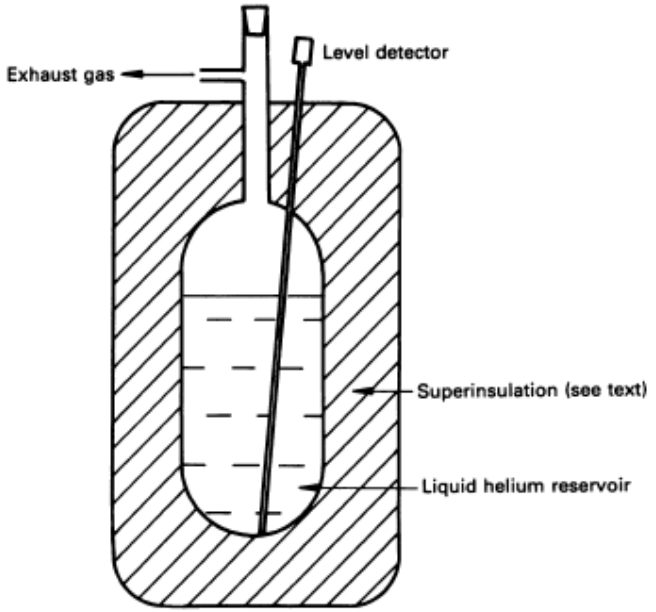
Reaching Low Temperature

Liquification of Helium

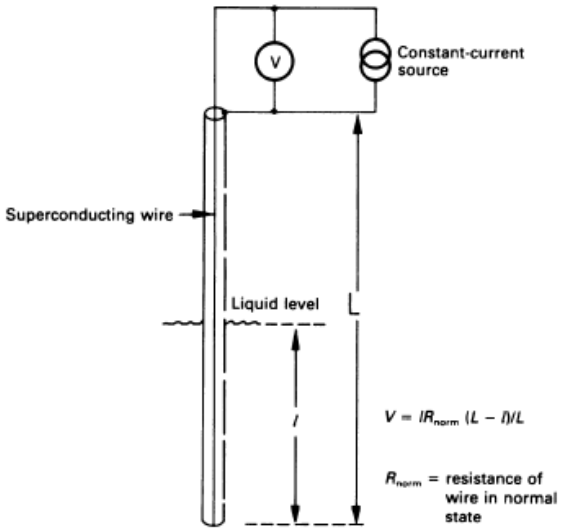


It is based on Joule-Kelvin expansion

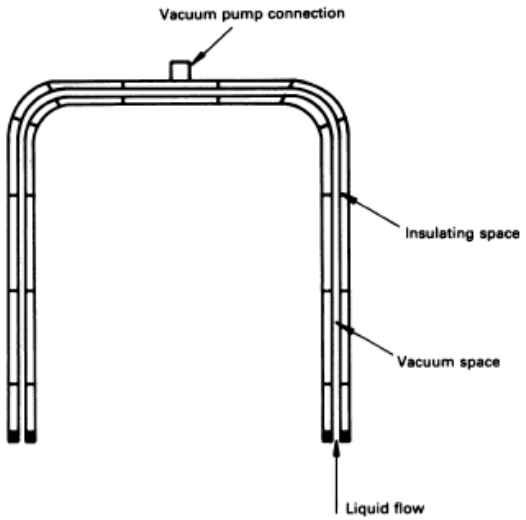


Schematic diagram of a modern stainless steel superinsulated helium Dewar

Helium liquefier and gas handling systems



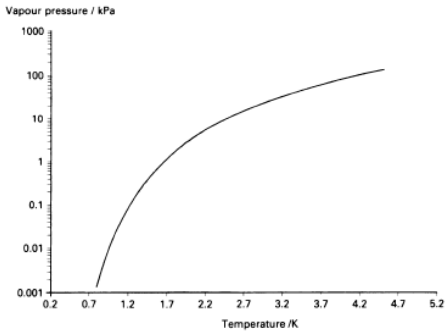
Superconducting liquid-helium depth sensor



Insulated helium transfer tube or syphon

Helium-4 Cryostats

Cryostats with Variable Temperature for 1.3K ≤ *T* ≤ 4.2K



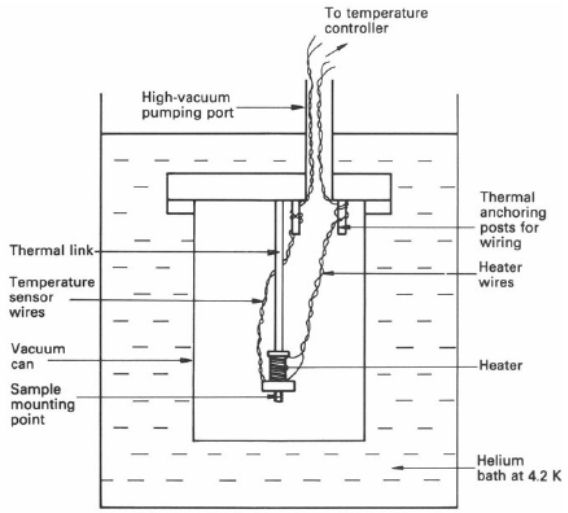
Boiling temperature of liquid 4He as a function of vapour pressure

With a sufficiently powerful pump, Temperatures down to about 0.75 K should be theoretically possible. In practice, it is possible to obtain temperatures between 4.2 and about 1 K.

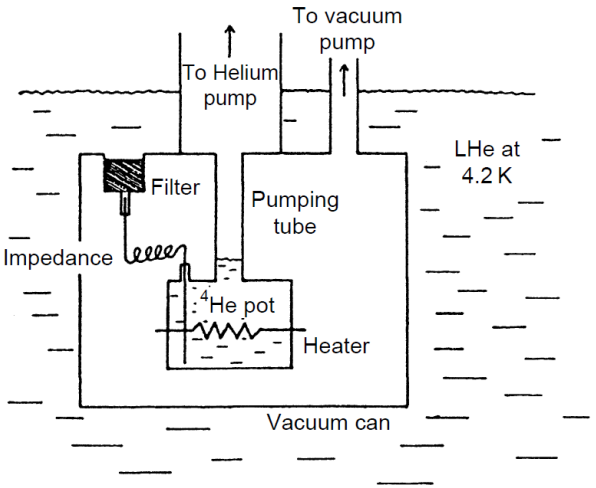
Cooling to below the A.-point, 2.17 K, where liquid 4 He becomes superfluid, is difficult. Surfaces in contact with superfluid are covered with a helium film a few atoms thick.

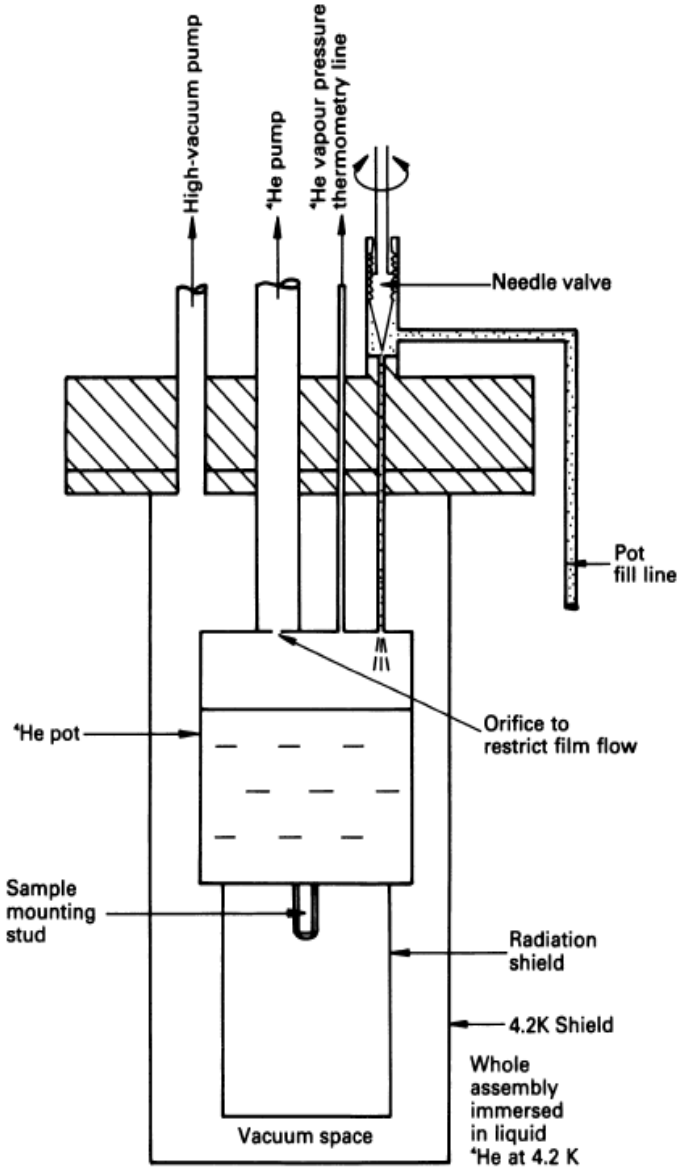
This flows rapidly until it reaches warmer parts of the apparatus and then evaporates, thus increasing the amount of gas the pump has to cope with.

The film flow can be reduced by reducing the size of the orifice through which it must flow (the thickness of the film being constant), but this also introduces a restriction in the gas flow, so limiting the cooling power of the system.



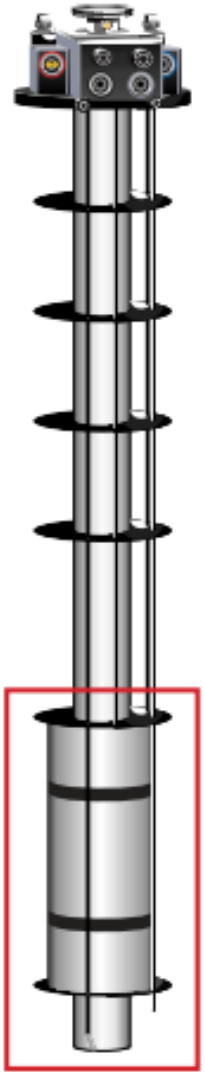
Experimental arrangement for obtaining sample temperatures above 4.2 K

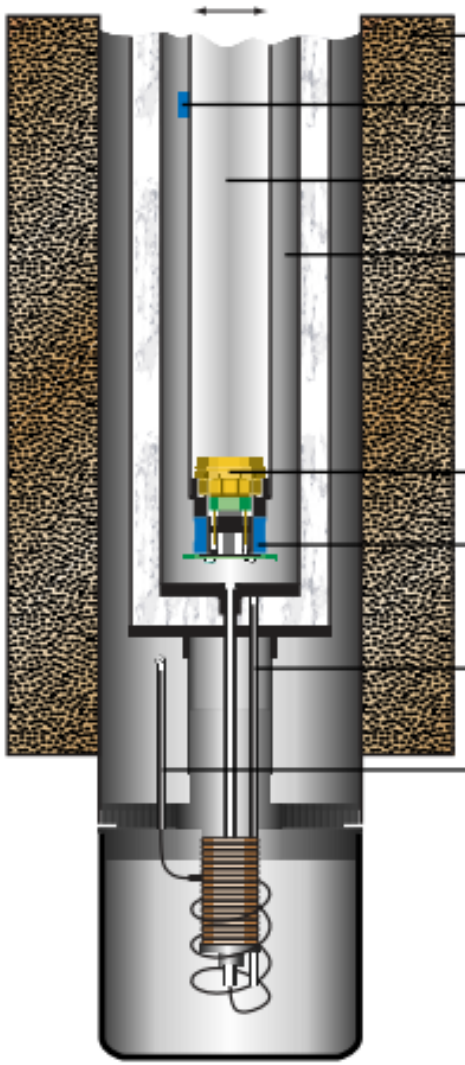




System for obtaining temperatures between 4.2 K and about ~ 1.3 K by pumping on a pot of liquid 4He

PPMS Probe





Magnet

Thermometer

Sealed sample space

Cooling annulus

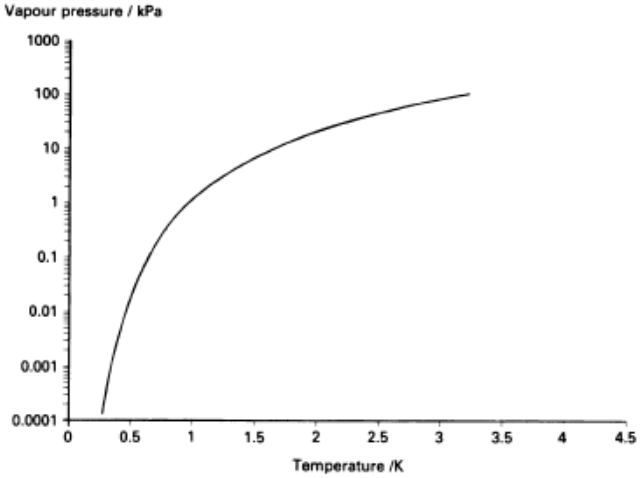
Puck

Heaters and thermometers (2)

Dual impedance system

- Multiple thermometers accurately monitor while heaters quickly control the system temperature.

Helium-3 Cryostats

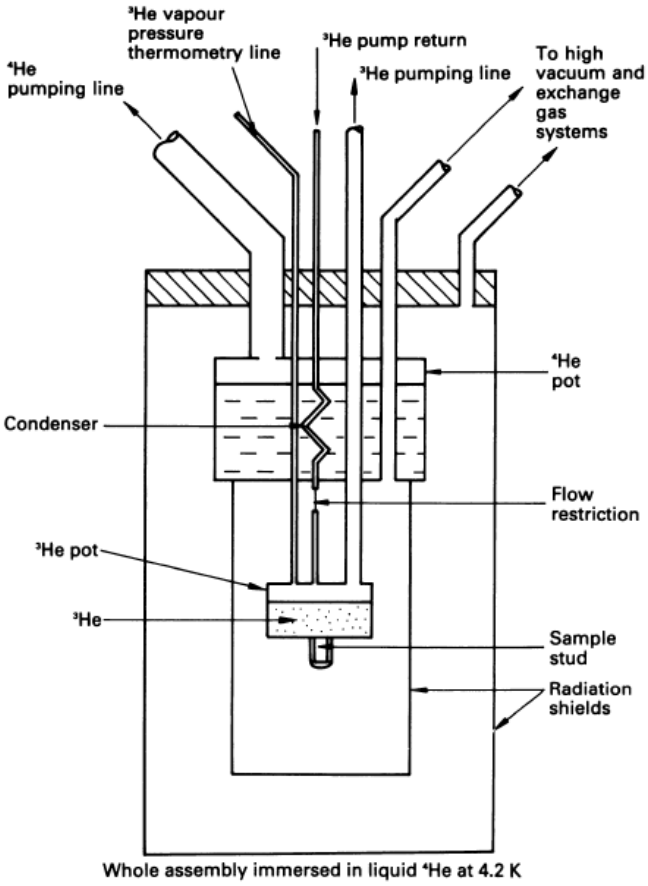


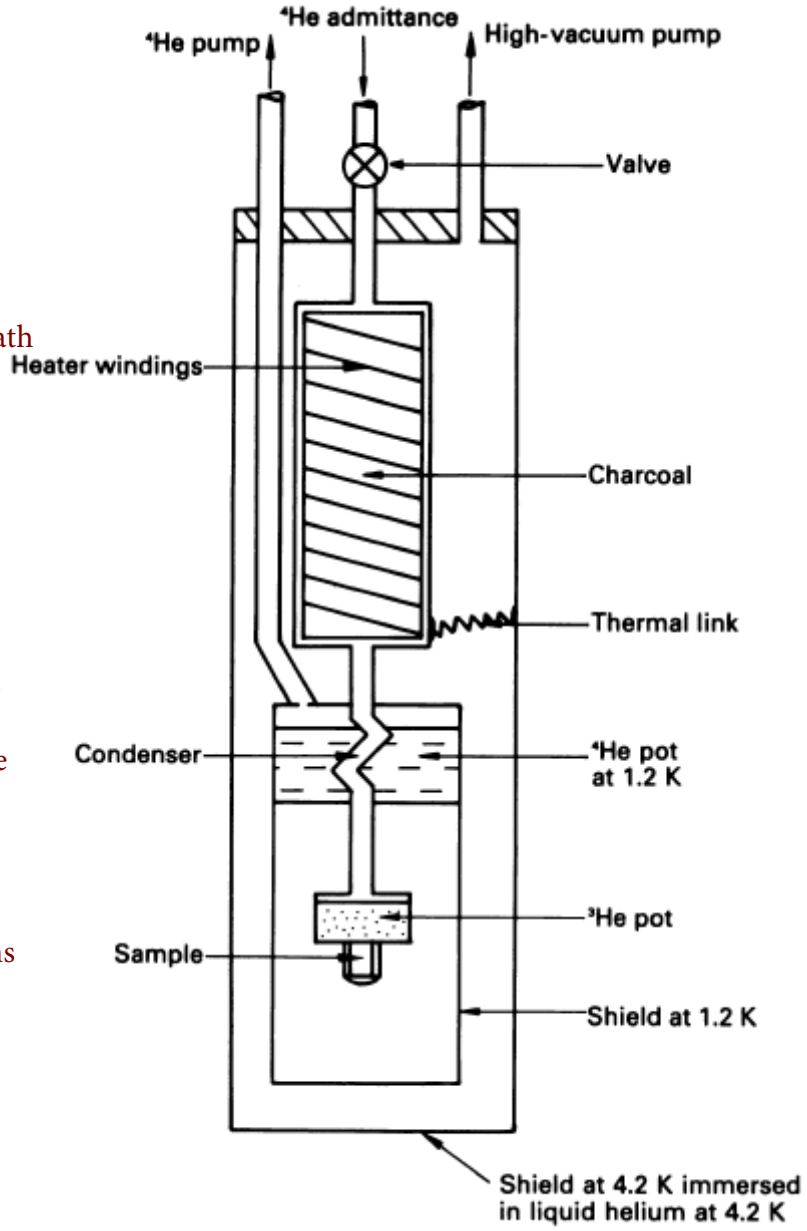
Temperatures down to about 0.3 K may be obtained by pumping on a bath of 3He.

There is no superfluid film flow to contend with.

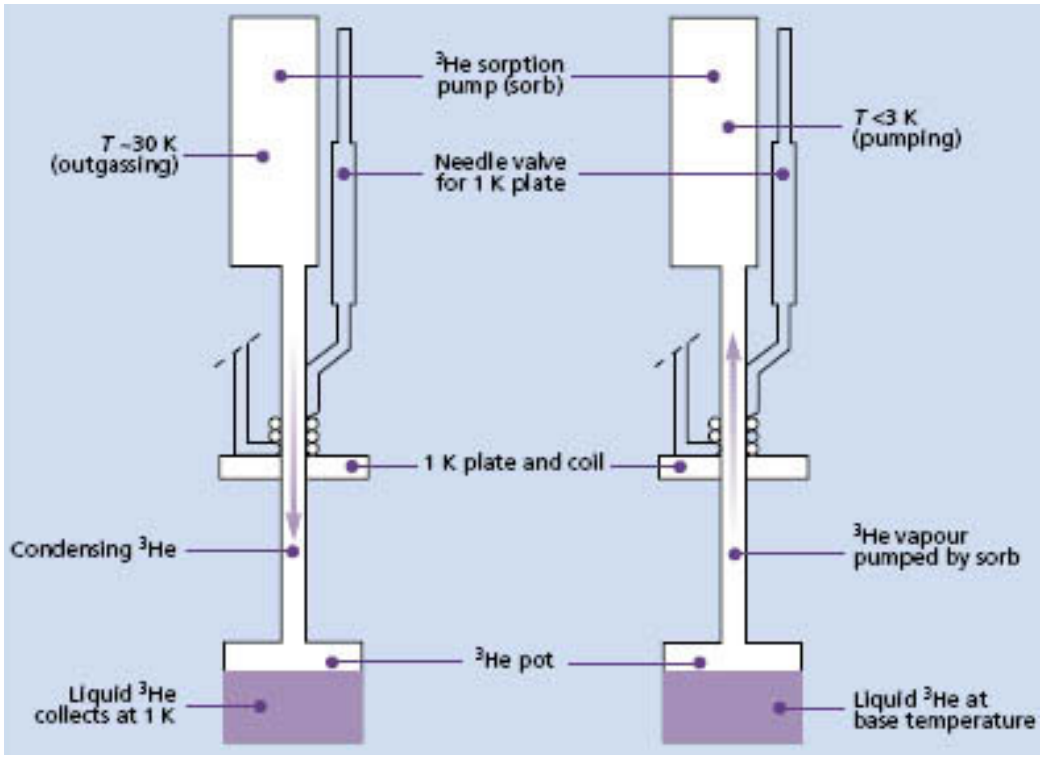
However, 3He is very expensive. It is produced by the radioactive (~-) decay of tritium, which has a half-life of 12.3 years.

Consequently, 3He cryogenic systems are designed to work with just a few cubic centimetres of iquid.





Oxford Instrument Heliox System



In phase 1, the 3He gas, cryo-pumped by the cryogenic pump (sorb), is free to condense in the 1 K plate region of the central tube and runs down to cool the 3He pot and sample. After approximately 20 minutes most of the gas has been condensed and the 3He pot is nearly full of liquid 3He at approximately 1.5 K.

In phase 2, the sorb is cooled to below 10 K and begins to reduce the vapour pressure above the liquid 3He. The sample temperature cools to its base temperature.

Helium-3 Cryostats





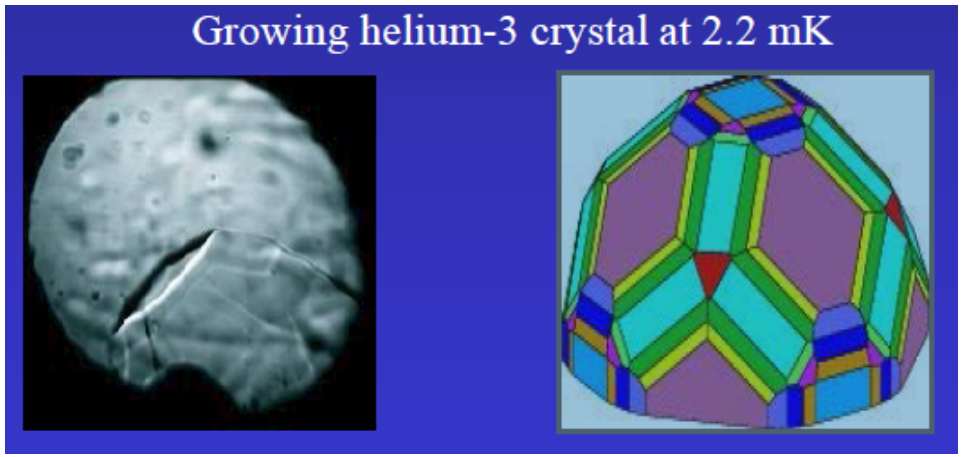


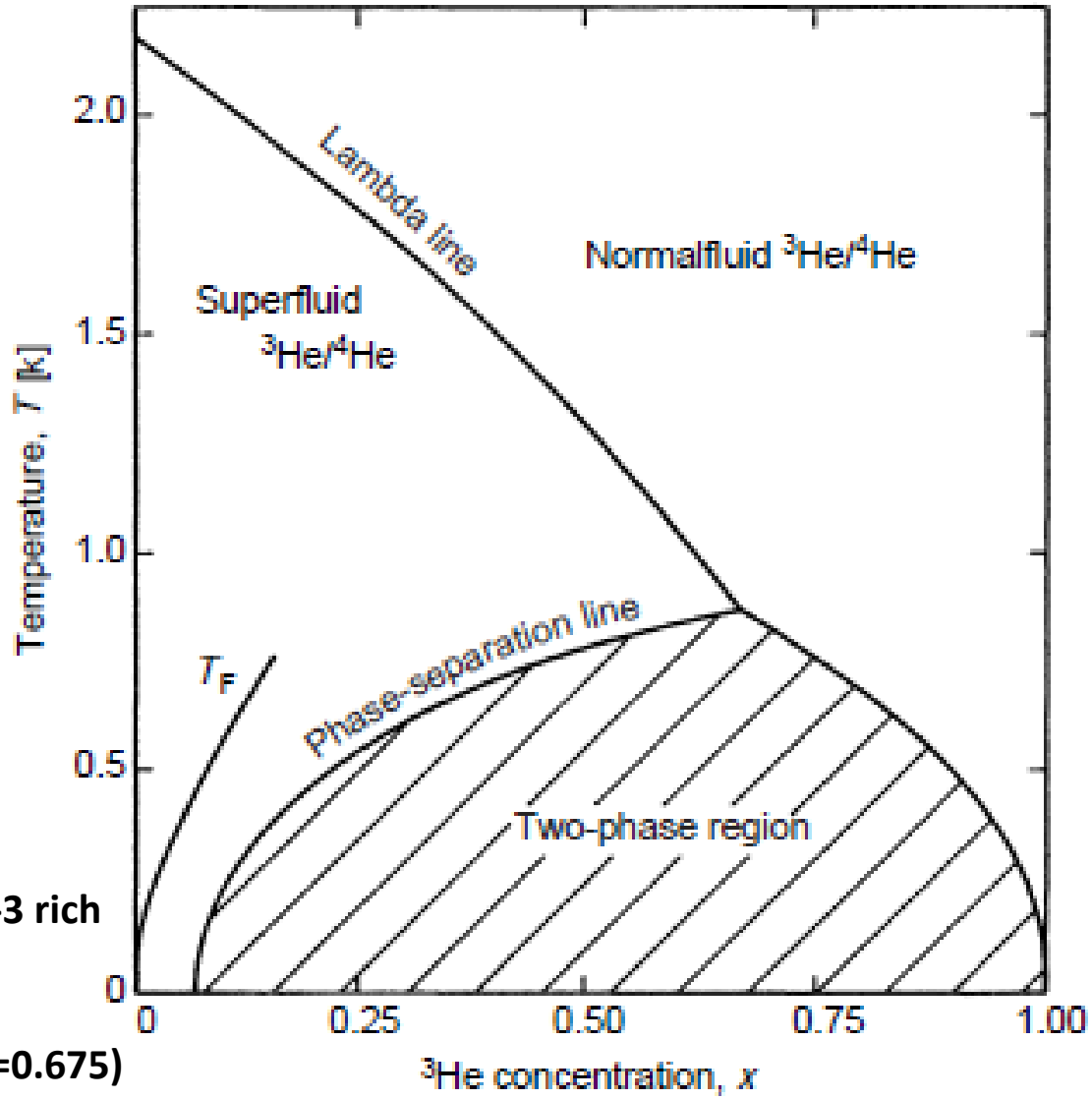

How to go below 300 mK ??

He-3 nucleus has spin 1/2, Fermion

Additional spin entropy

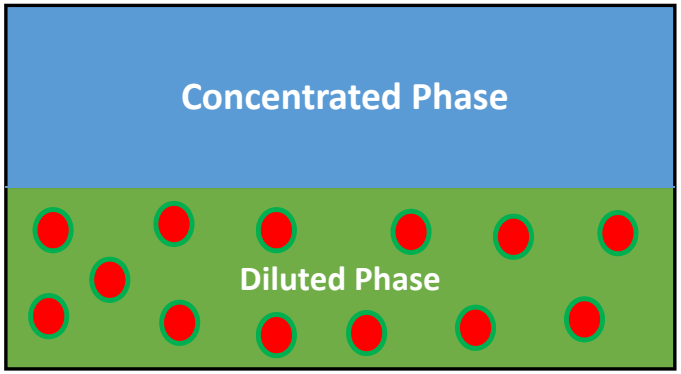
several superfluid phases





Mixture of He-3 and He-4

- **He-3 He-4 Phase separation of the mixture into He-3 rich and He-3 poor phases, but not pure He-3 and He-4**
- **Phase separation starts below T=0.867 K (max at X=0.675)**



^{3}He Only

^{4}He + a little of ^{3}He

At T > 0.8 K : the liquids are miscible (like water and alcohol)

At T < 0.8 K : separation in 2 phases (like water and oil)

In the lower phase : 3-He diluted in 4-He = diluted phase

The concentration of 3-He depends on temperature

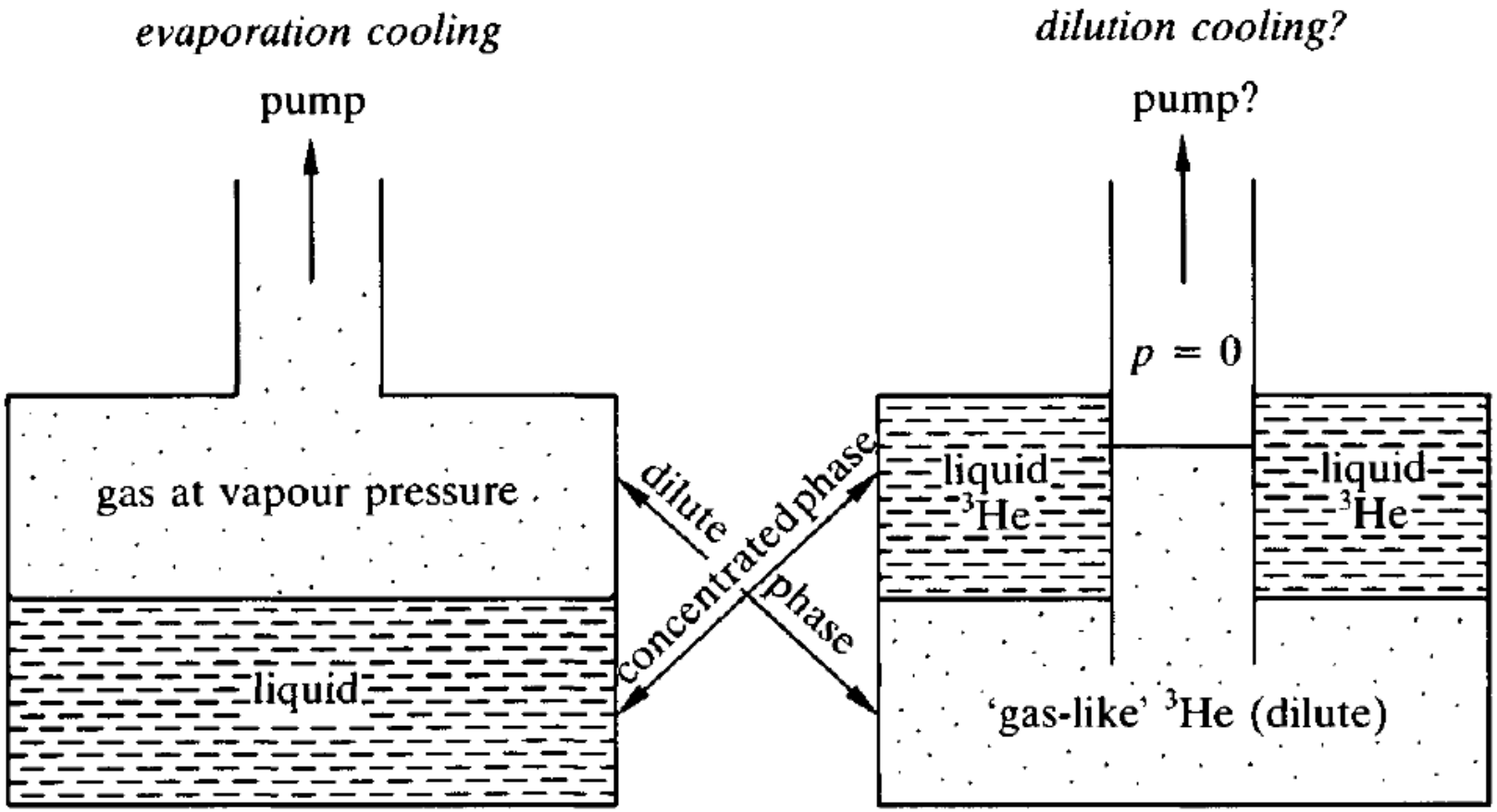
0.5 K ~ 20 % 3-He , T ~ 0 K ~ 6.4 % 3-He

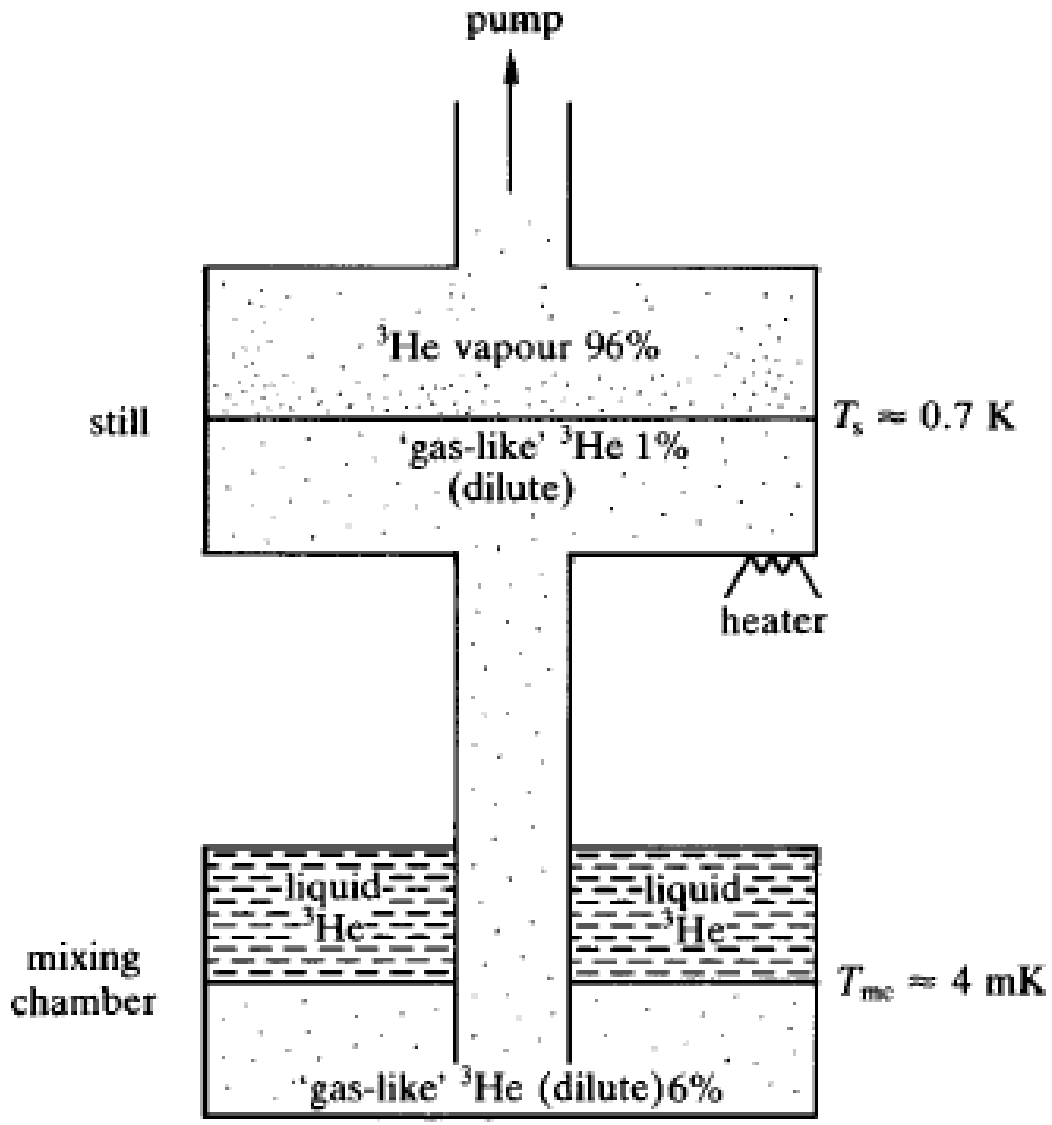
Even at low temperatures, there is a finite solubility of 3-He in 4-He

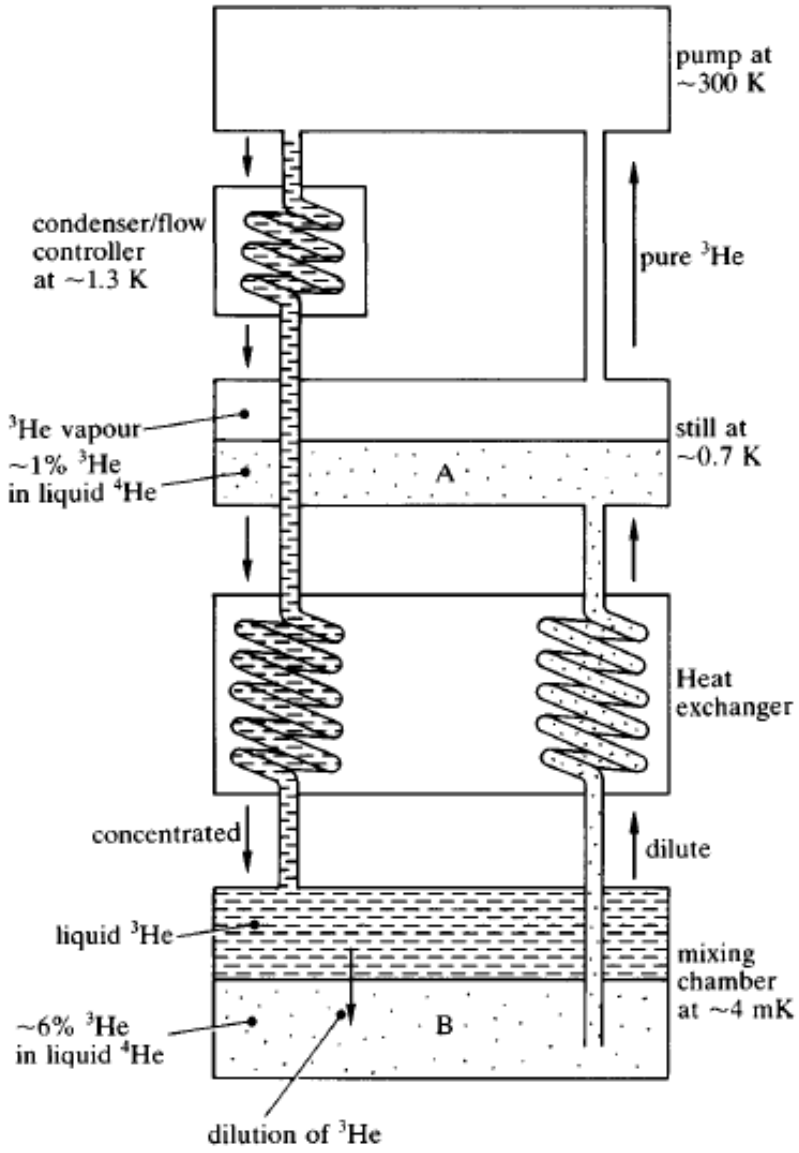
In the upper phase : concentrated 3-He (no 4-He)

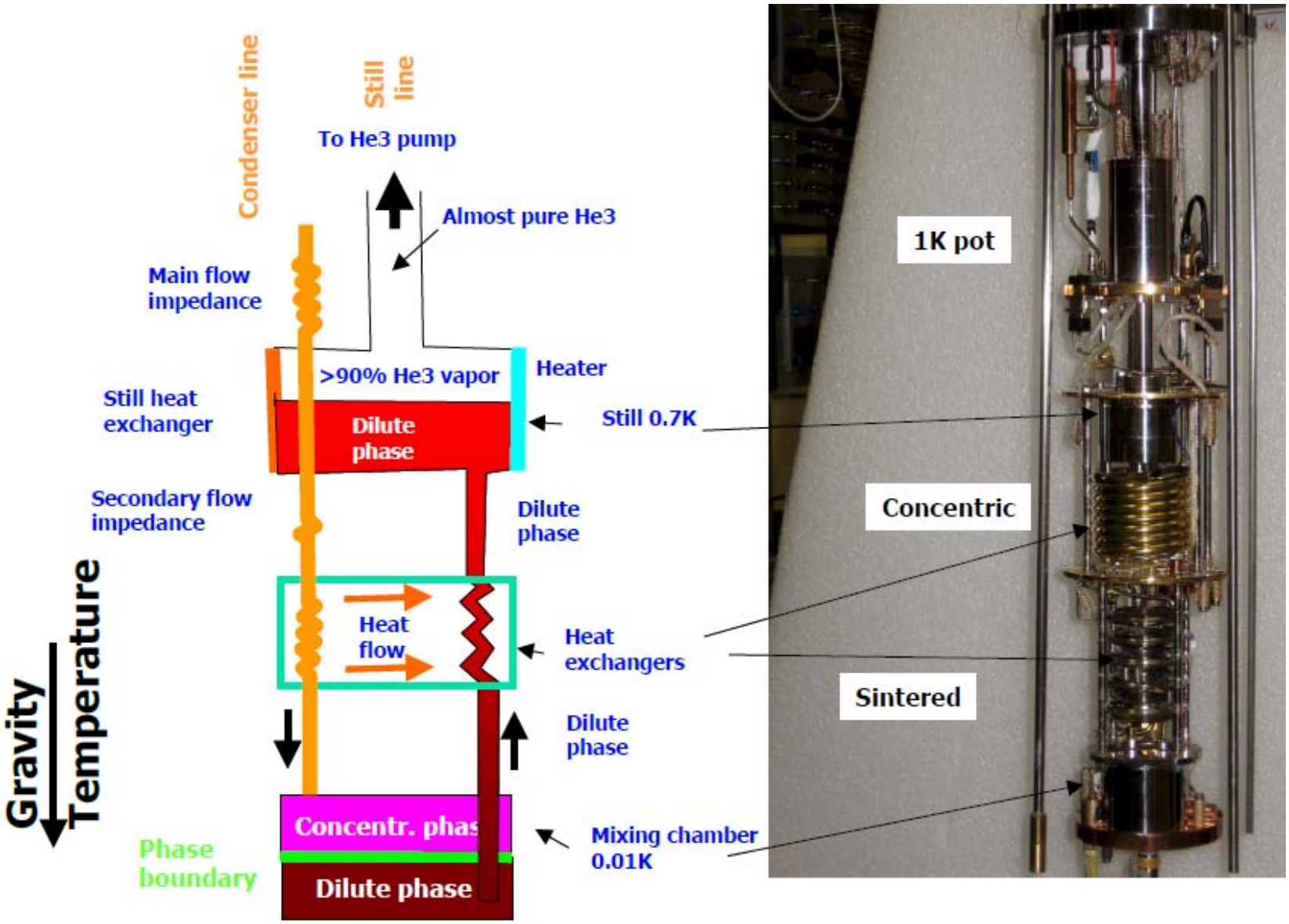
4-He is superfluid in the diluted phase

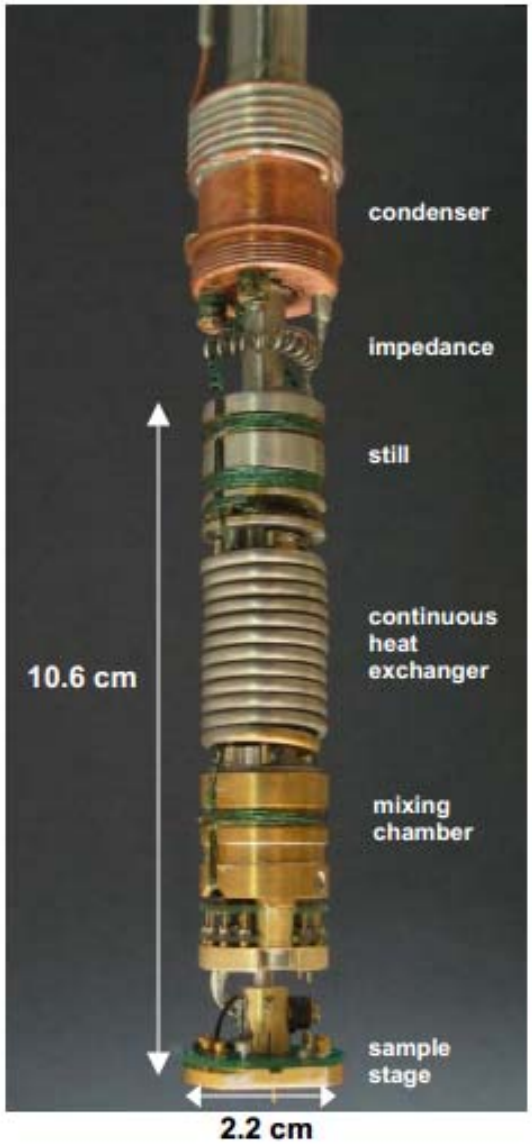
Dilution Refrigerator

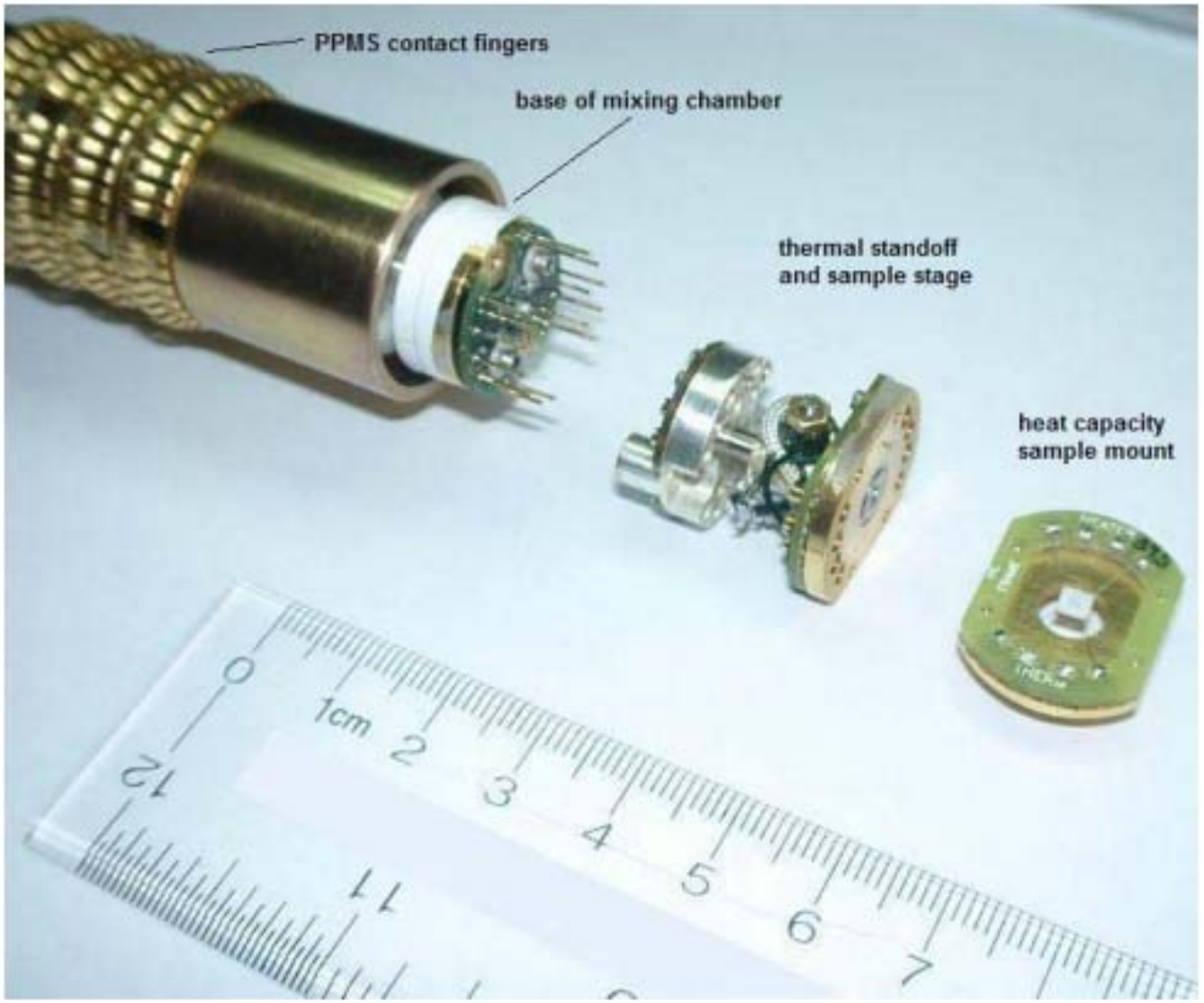



pump
still
^{3}He vapour 96%
'gas-like' ^{3}He 1%
(dilute)
$T_s \approx 0.7$ K
heater
mixing
chamber
liquid
^{3}He
liquid
^{3}He
$T_{mc} \approx 4$ mK
'gas-like' ^{3}He (dilute)6%

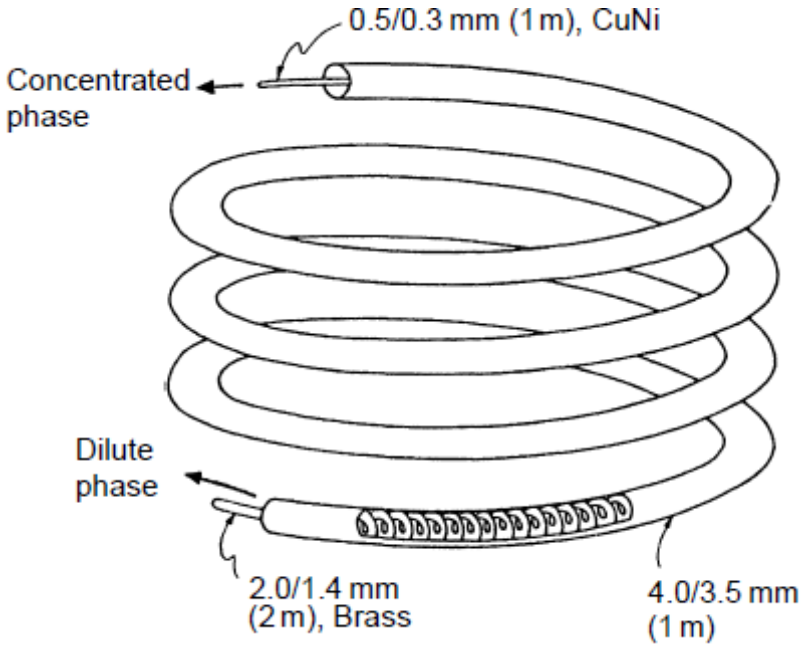

pump at
~300 K
condenser/flow
controller
at ~1.3 K
pure ^{3}He
^{3}He vapour
~1% ^{3}He
in liquid ^{4}He
A
still at
~0.7 K
Heat
exchanger
concentrated
dilute
liquid ^{3}He
mixing
chamber
at ~4 mK
~6% ^{3}He
in liquid ^{4}He
B
dilution of ^{3}He

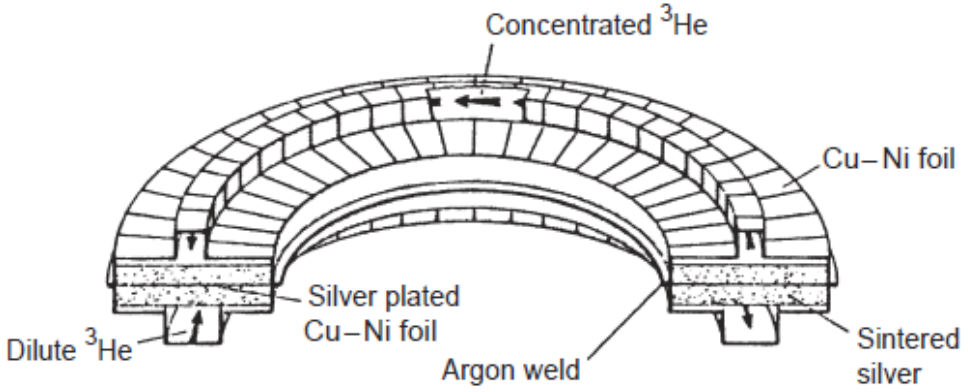

Condenser line
Still line
To He3 pump
Almost pure He3
Main flow impedance
>90% He3 vapor
Heater
Still heat exchanger
Dilute phase
Still 0.7K
Secondary flow impedance
Dilute phase
Heat flow
Heat exchangers
Dilute phase
Gravity
Temperature
Concentr. phas
Phase boundary
Dilute phase
Mixing chamber 0.01K
1K pot
Concentric
Sintered


condenser
impedance
still
continuous heat exchanger
mixing chamber
sample stage
10.6 cm
2.2 cm

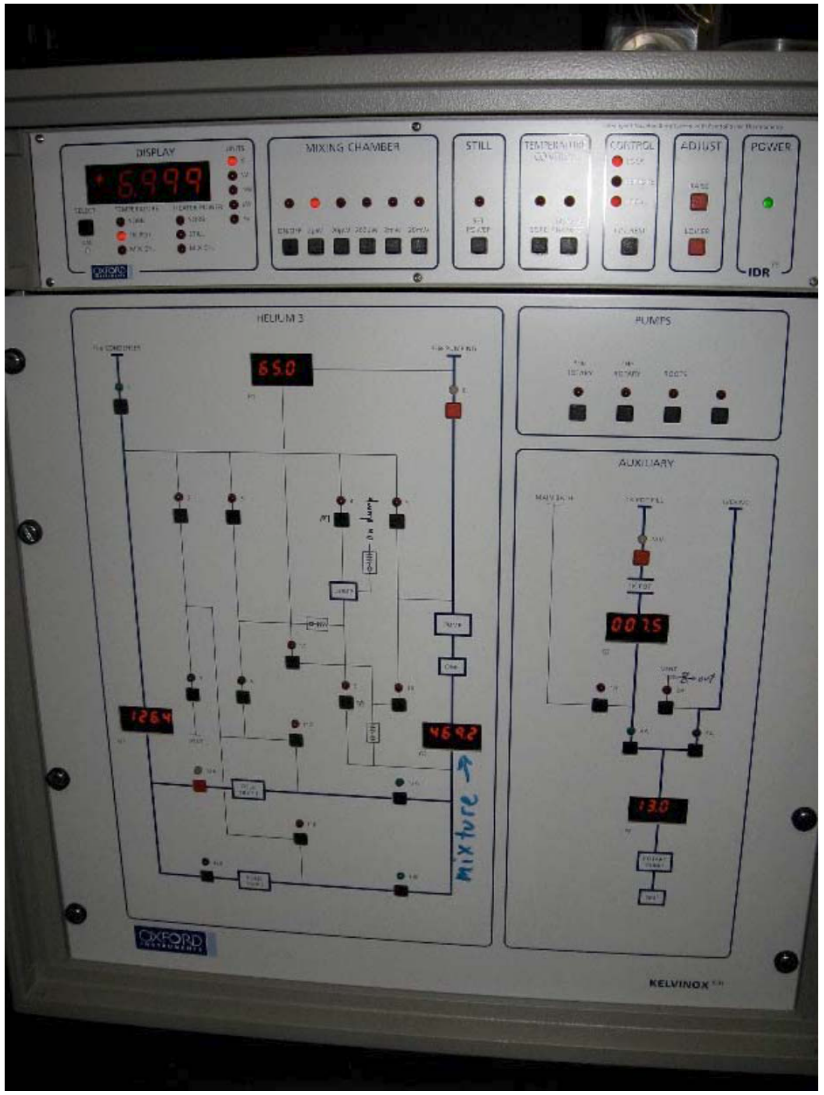

PPMS contact fingers
base of mixing chamber
thermal standoff and sample stage
heat capacity sample mount
0
1cm
2
3
4
5
6
7

Heat Exchanger





http://www.magnet.fsu.edu/education/tutorials/tools/dilutionfridge.html


DISPLAY
6999
MIXING CHAMBER
STILL
TEMPERATURE
CONTROL
ADJUST
POWER
IDR
HELIUM 3
PUMPS
AUXILIARY
65.0
12.64
46.92
00.75
1.30
Mixture
OXFORD
KELVINOX


PERIODIC TABLE OF THE ELEMENTS


PFEIFFE
SOGEVAC


EDWARDS

